

Network management, performance and security

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Book chapters

Forouzan: 19.1 (NAT), 21.1 (DHCP), 21.2, 24.2,
31, 32

Kihl: 10.2-10.3, 11

IP address allocation

Each host that connects to a network must have an IP address. A network manager has two ways to allocate IP addresses within the network:

- **Static allocation:** When a host first connects to the network it is manually given a permanent address.
- **Dynamic allocation:** There is a pool of available addresses that are dynamically allocated to all active hosts.

Dynamic Host Configuration Protocol (DHCP)

DHCP provides both static and dynamic address allocation.

It has one database that statically binds physical addresses to IP addresses.

It has a second database with a pool of available IP addresses.

A DHCP client (host) sends a request to the DHCP server, and is either allocated a static address from the first database or a temporary address from the second database.

Network Address Translation (NAT)

NAT enables a private network to have a large set of internal addresses and only a small set of external addresses.

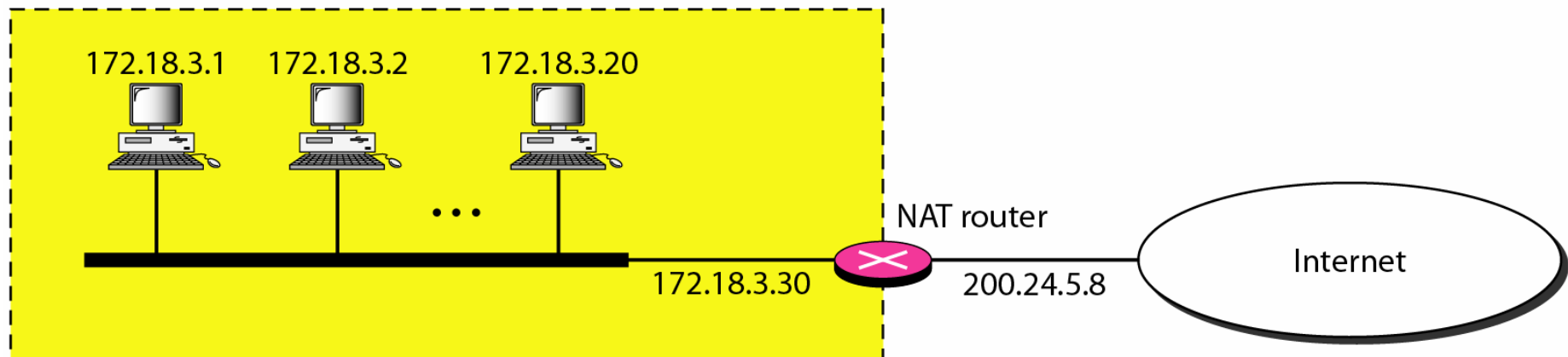
The Internet authorities have reserved three sets of addresses as **private addresses**:

<i>Range</i>			<i>Total</i>
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	172.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}

A NAT implementation

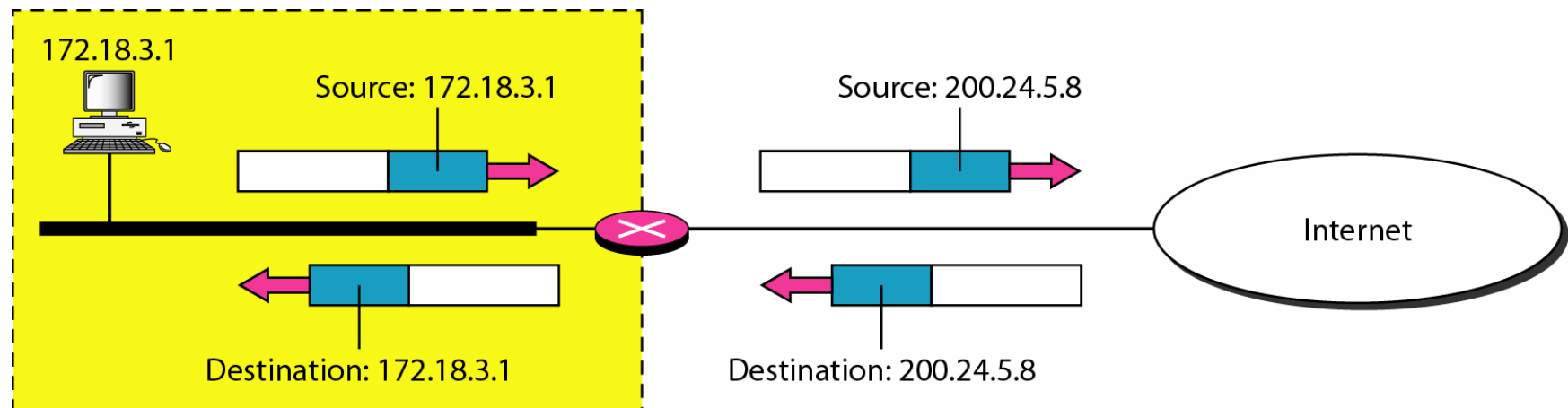
The private network must have only one single connection to the global Internet through a router that runs NAT software:

Site using private addresses



Address translation

All the outgoing packets go through the NAT router, which replaces the source address in the packet with the global NAT address. For incoming packets, the destination address is replaced.



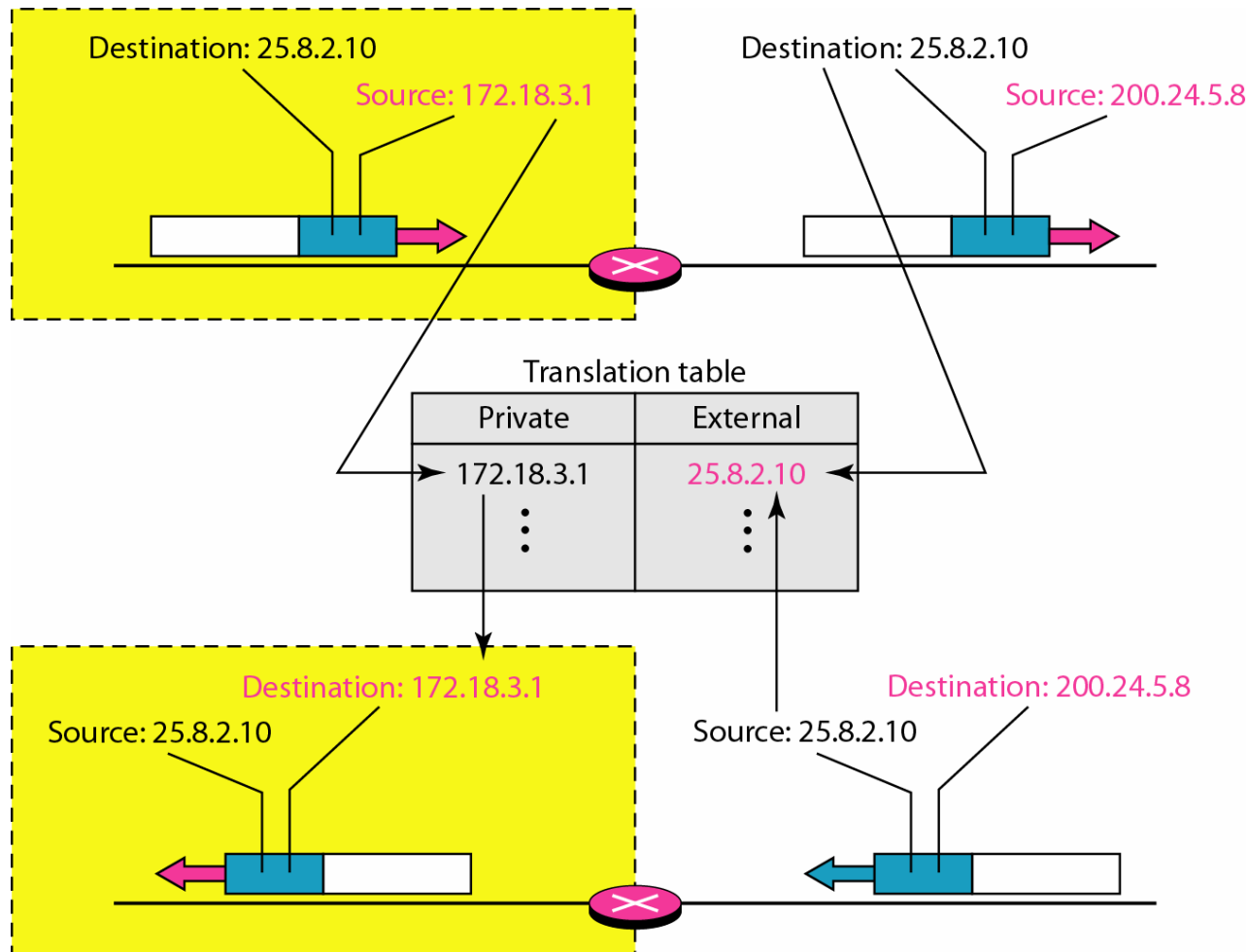
Translation table

The correct destination address for an incoming packet is found in a translation table that maps an external source address with a private destination address.

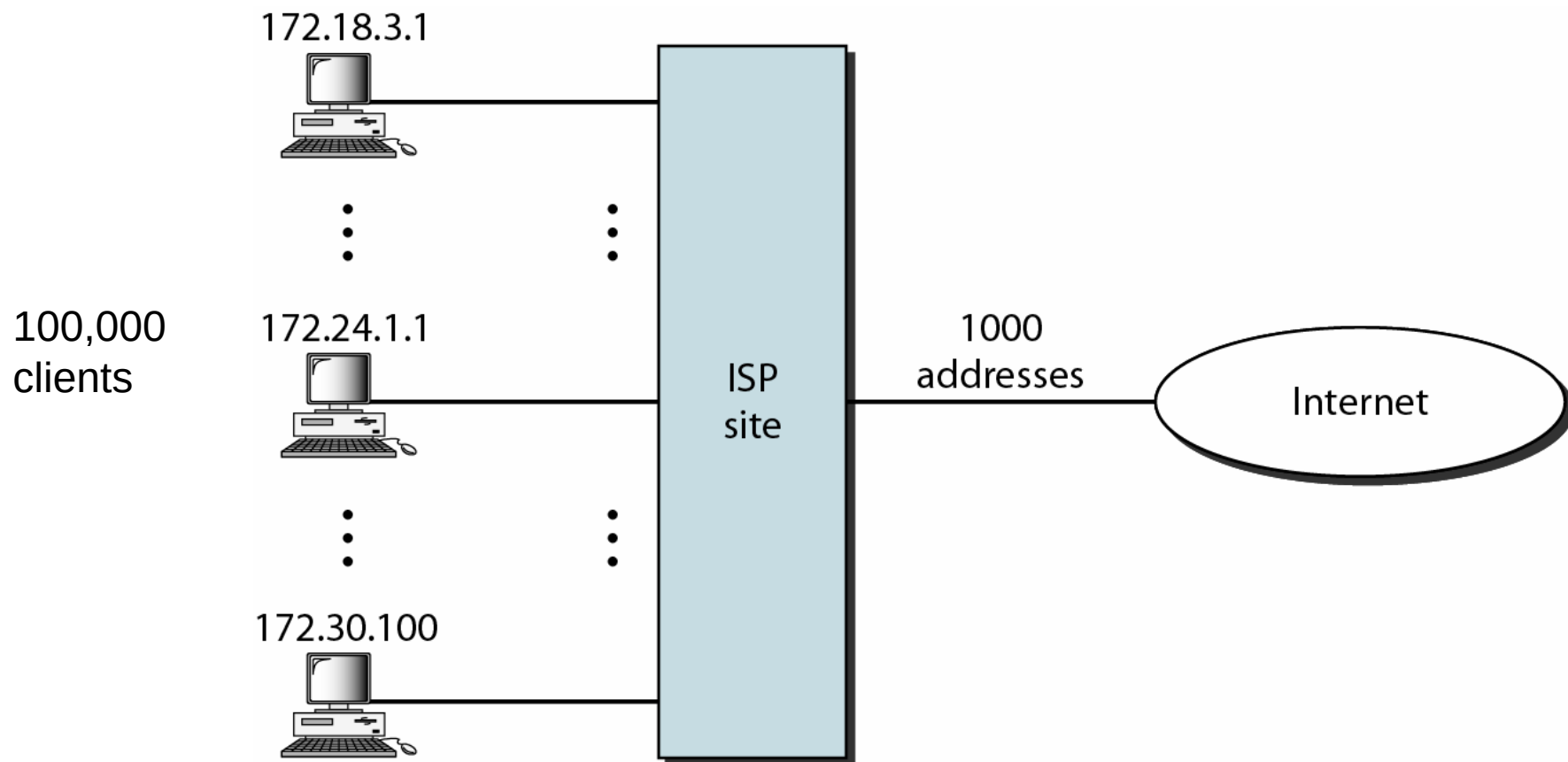
In order for this to work, the communication must always be initiated by the host in the private network.

The NAT router makes a note of the destination address in the translation table whenever a host in the private network sends a packet to an external address.

Address translation, example



Pool of addresses



Network Performance

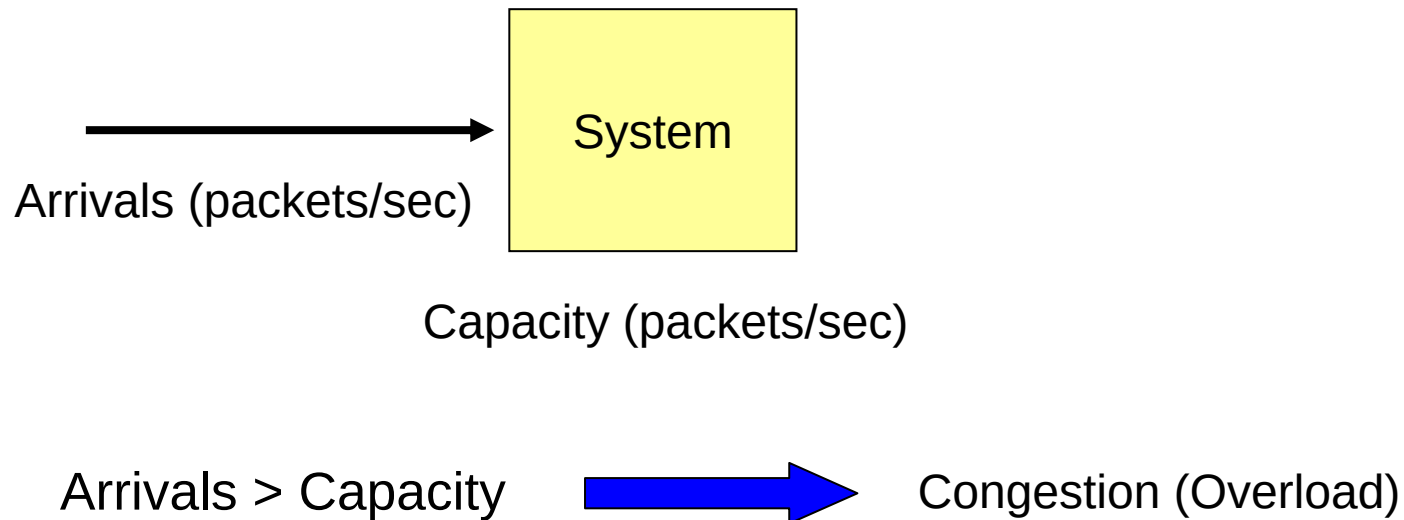
A good network performance is crucial for most applications. Two factors of a network or system that influence the performance are:

- Delay
- Throughput

Both these factors are dependent on the **load** on the network or system.

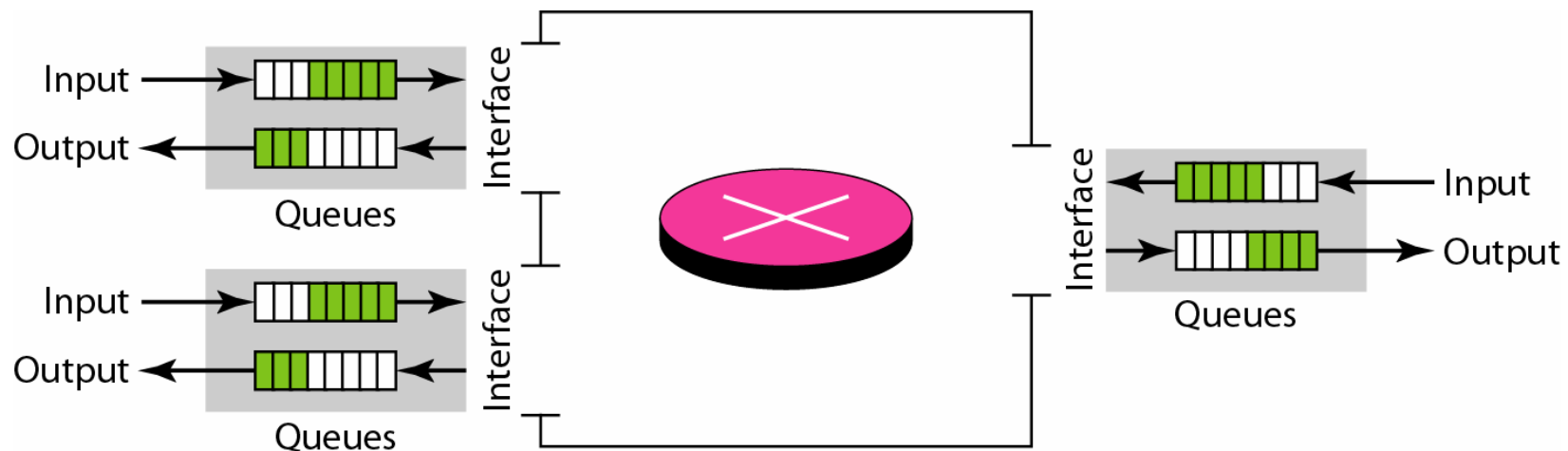
Congestion

Congestion (Overload) occurs when the number of packets sent to a system is greater than the number of packets the system can handle.



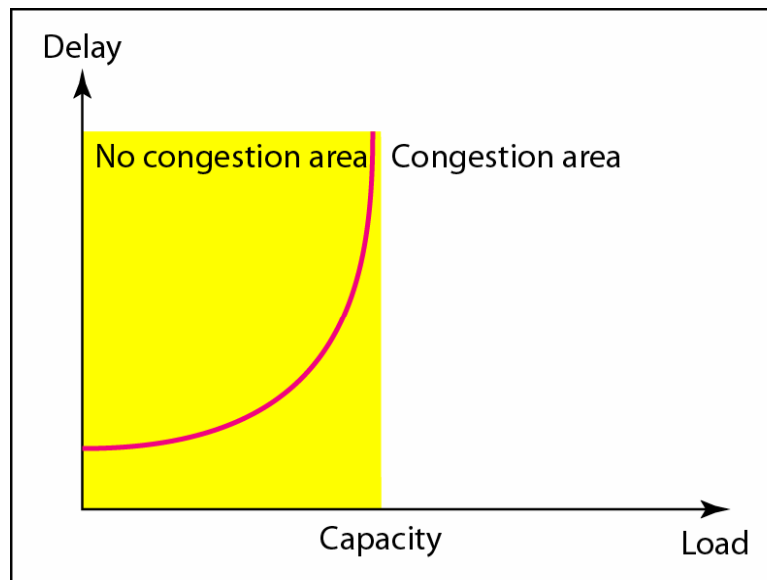
Queuing delay

All routers and servers have buffers where jobs are stored in wait for service. When the load increases, the buffers fill up, and the **queuing delay** increases. Finally, packets will be dropped due to full buffers.



Delay as a function of load

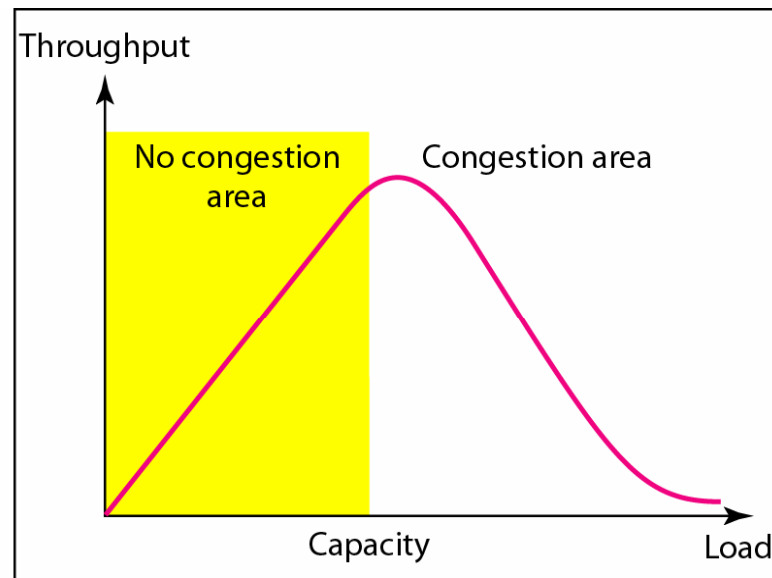
The delay will increase exponentially to the load.
This can be shown with [queuing theory](#).



a. Delay as a function of load

Throughput as a function of load

During overload, the throughput declines sharply.
Long delays casues retries from hosts or routers.
All dropped packets also cost capacity.

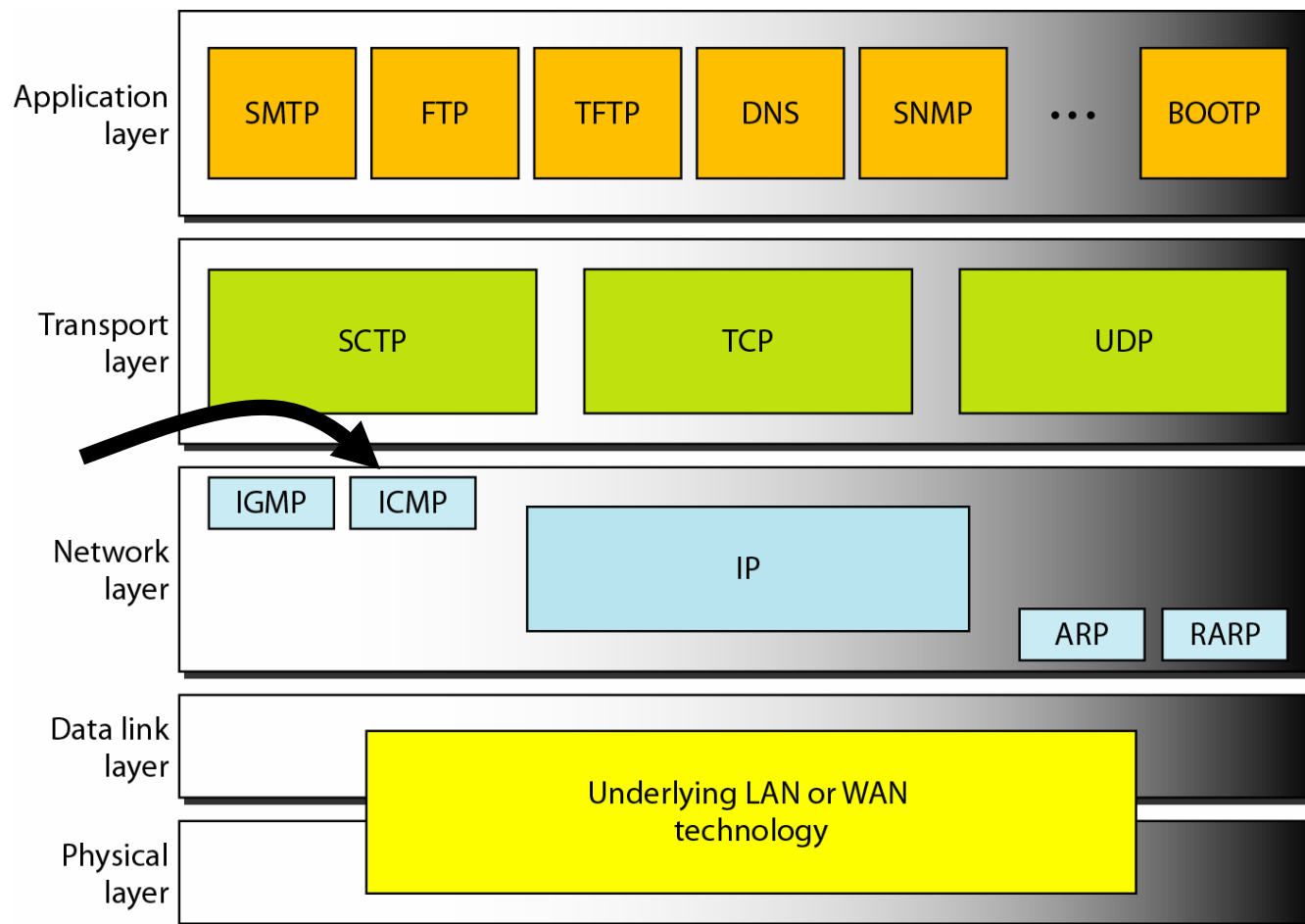


b. Throughput as a function of load

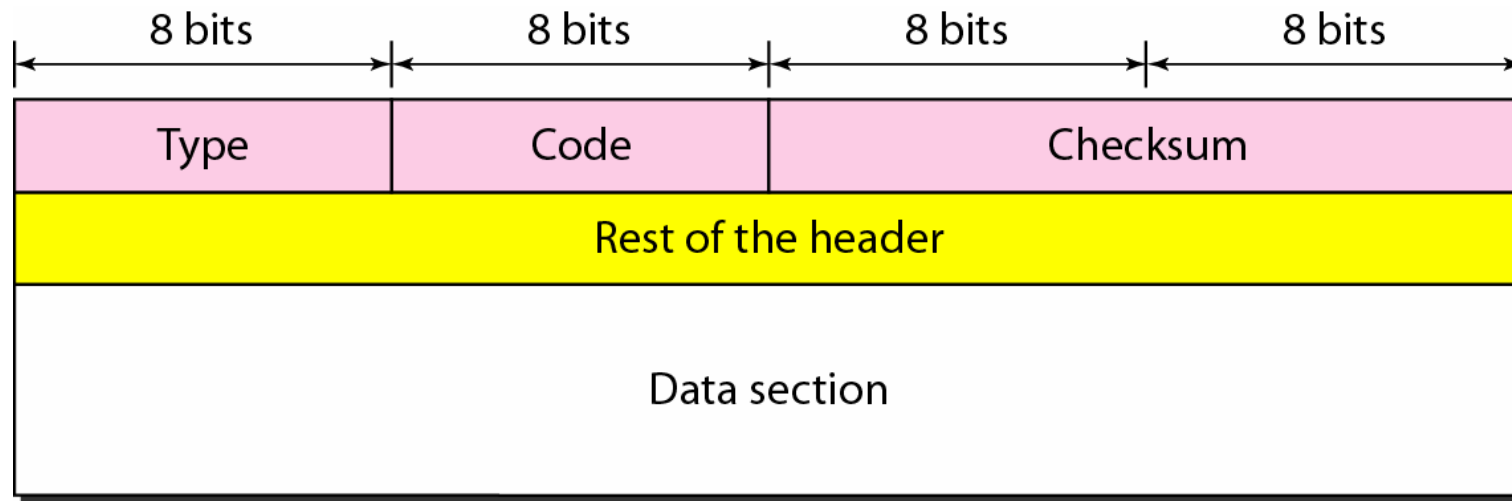
Internet Control Message Protocol (ICMP)

- The IP protocol has no error-reporting or error-correcting mechanism. The IP protocol also lacks a mechanism for host and management queries.
- The [Internet Control Message Protocol \(ICMP\)](#) has been designed to compensate for the above two deficiencies.
- It is a companion to the IP protocol.

ICMP in the TCP/IP stack



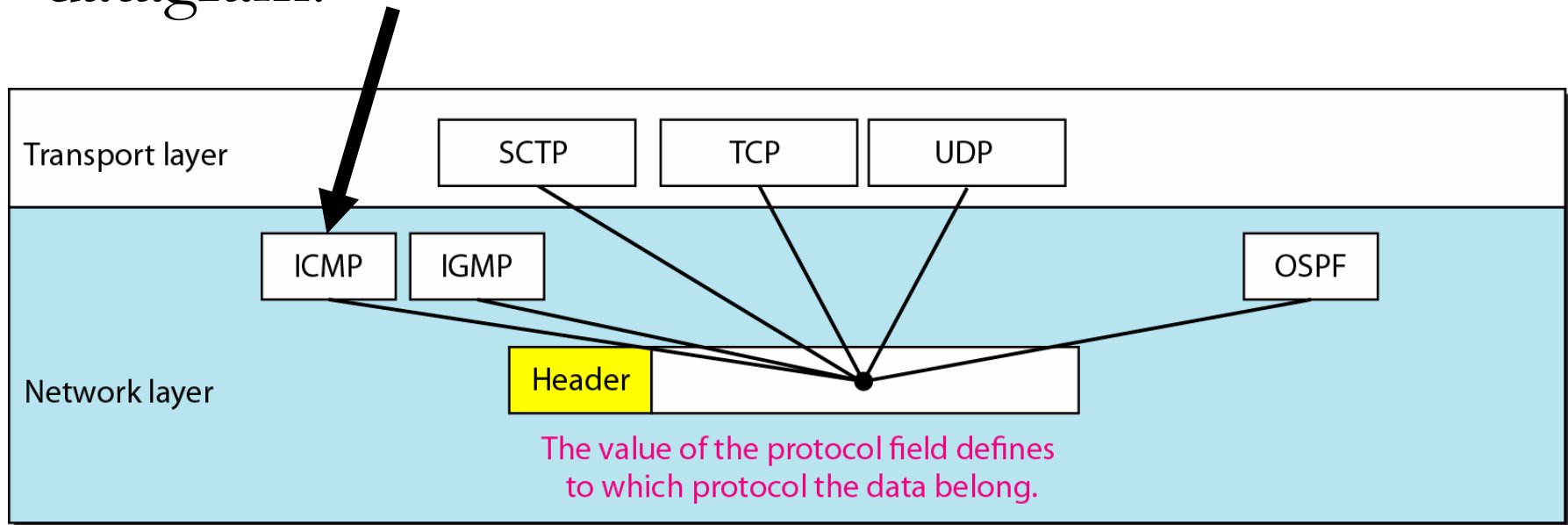
ICMP message



Two types of messages:
Error-reporting messages
Query messages

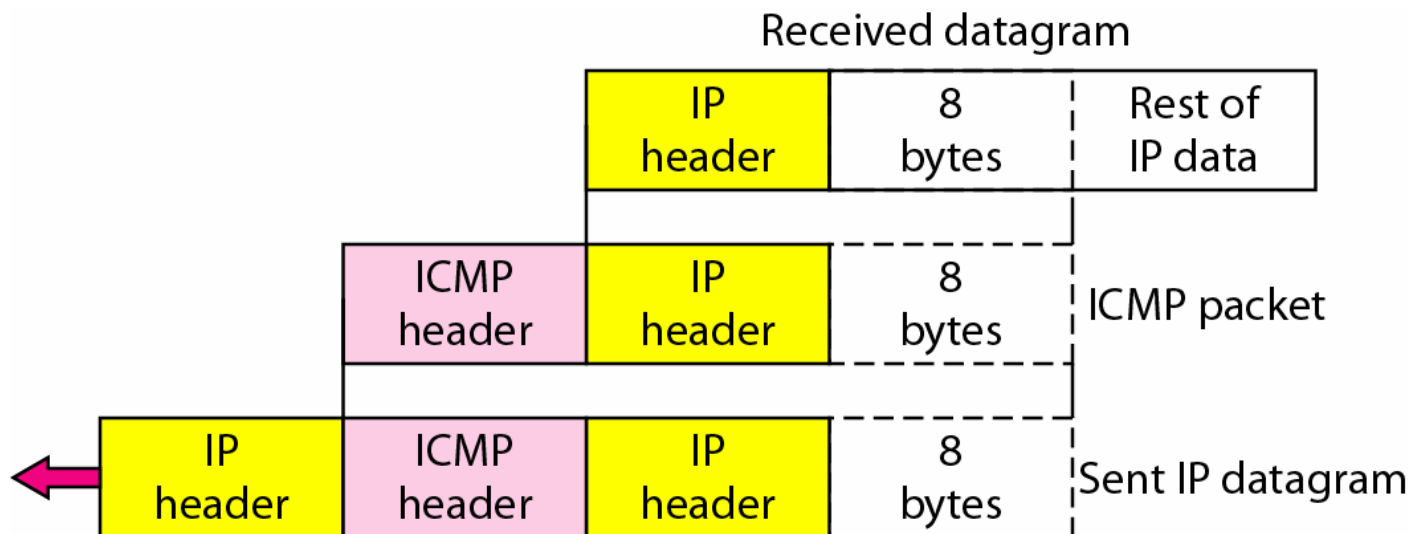
Encapsulation

The ICMP message is encapsulated in an IP datagram:



Error-reporting

When an error in the IP datagram *delivery* is detected, ICMP reports the error to the source of the IP datagram. The error message includes the IP header and first 8 bytes of the original datagram.



Some Error reporting messages

- **Destination unreachable**: A router cannot route a datagram or a host cannot deliver a datagram.
- **Source quench**: A datagram is discarded due to congestion.
- **Time exceeded**: A datagram has been discarded since its TTL value has reached 0.
- **Redirection**: A host has the wrong default router, and needs to update its routing table.

Some ICMP Query messages

- **Echo-request and Reply**: Is used to determine if two systems (hosts or routers) have connection on the IP level.
- **Timestamp request and reply**: Is used to determine the round-trip time (RTT) between two systems.
- **Router-Solicitation and Advertisement**: Is used by a host to find the routers that are connected to its network.

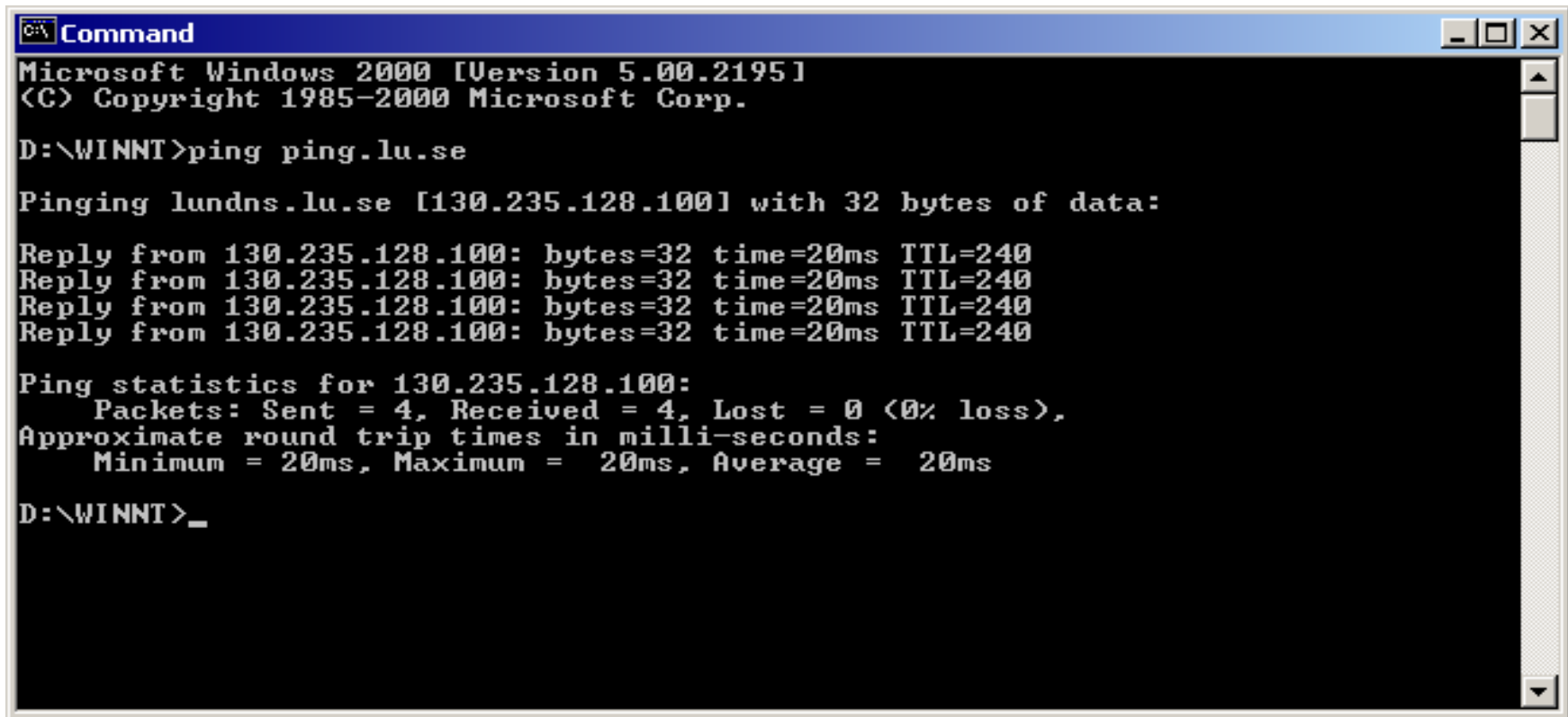
Debugging Tools

There are several tools that can be used in the Internet for debugging. Some of them are:

- Ping
- Traceroute
- Nslookup/host
- “Sniffer” tools

Ping

The Ping program uses ICMP echo-request and reply messages to find information about a destination.

A screenshot of a Windows Command Prompt window. The title bar is blue and says "Command". The window has a black background with white text. The text shows the command prompt at "D:\WINNT>", the command "ping ping.lu.se", the output of the ping command showing four successful replies from 130.235.128.100 with 32 bytes of data, a time of 20ms, and a TTL of 240. It also shows ping statistics for 130.235.128.100: 4 packets sent, 4 received, 0 lost (0% loss), and approximate round trip times of 20ms minimum, maximum, and average. The prompt ends with "D:\WINNT>_".

```
Command
Microsoft Windows 2000 [Version 5.00.2195]
(C) Copyright 1985-2000 Microsoft Corp.

D:\WINNT>ping ping.lu.se

Pinging lundns.lu.se [130.235.128.100] with 32 bytes of data:

Reply from 130.235.128.100: bytes=32 time=20ms TTL=240
Reply from 130.235.128.100: bytes=32 time=20ms TTL=240
Reply from 130.235.128.100: bytes=32 time=20ms TTL=240
Reply from 130.235.128.100: bytes=32 time=20ms TTL=240

Ping statistics for 130.235.128.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 20ms, Average = 20ms

D:\WINNT>_
```


Traceroute

Traceroute (in UNIX) or Tracert (in Windows) can be used to trace the route of a packet from the source to the destination.

It manipulates the TTL field in the IP header and uses two ICMP messages: Time Exceeded and Destination Unreachable to find the route of a packet.

Traceroute, example

```
C:\ Command
D:\WINNT>
D:\WINNT>
D:\WINNT>
D:\WINNT>
D:\WINNT>
D:\WINNT>
D:\WINNT>tracert ping.lu.se

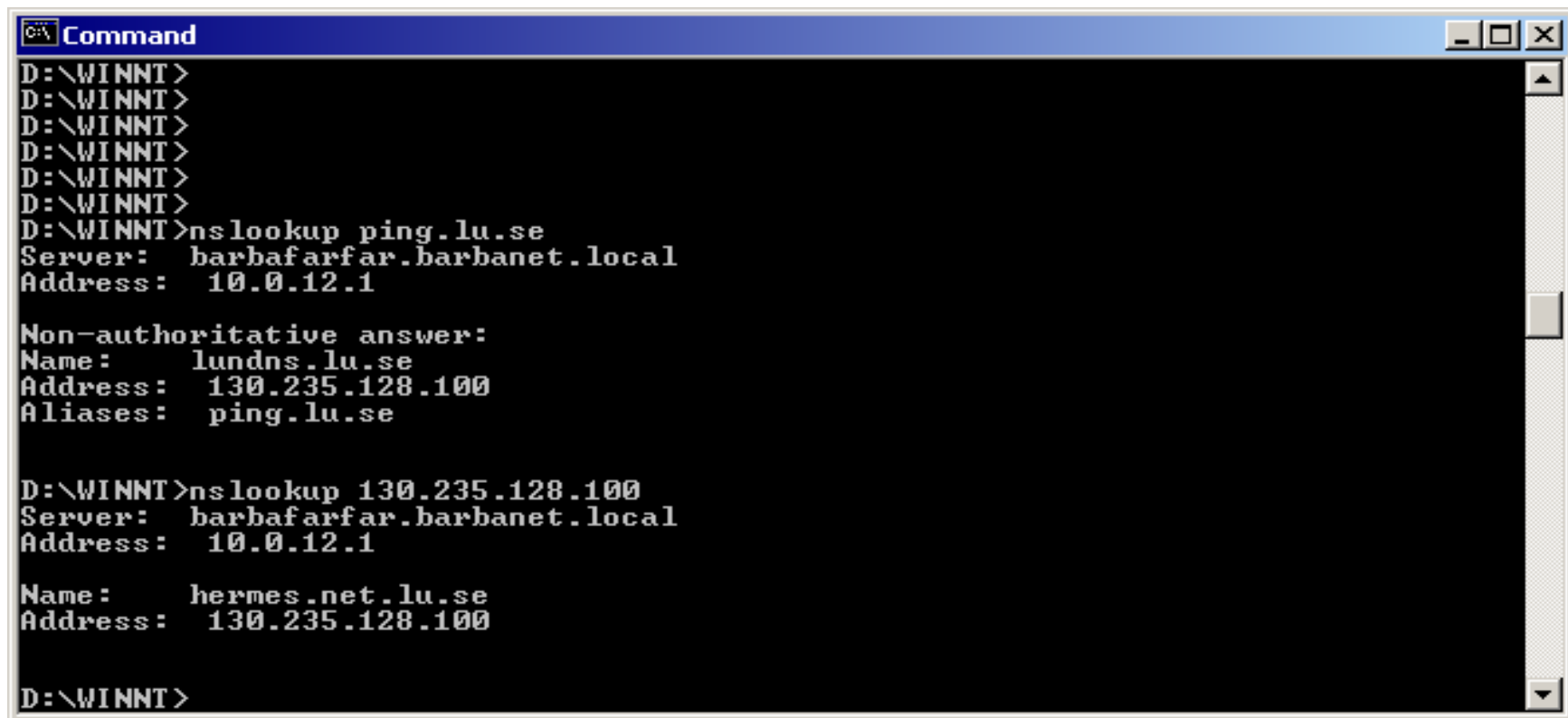
Tracing route to lundns.lu.se [130.235.128.100]
over a maximum of 30 hops:

  1  <10 ms    10 ms    <10 ms    barbafarfar.barbanet.local [10.0.12.1]
  2  <10 ms    10 ms    10 ms     gw-n1fls302o1100.telia.com [194.236.208.1]
  3  10 ms     10 ms    10 ms     10.0.111.1
  4  10 ms     10 ms    10 ms     217.211.120.187
  5  10 ms     10 ms    10 ms     m-b-c1-link.se.telia.net [81.228.72.108]
  6  10 ms     10 ms    50 ms     m-b-d1-link.se.telia.net [81.228.72.107]
  7  10 ms     10 ms    10 ms     malmo4-ge2.sunet.se [195.69.117.19]
  8  20 ms     60 ms    *         lu2-SRP1.sunet.se [130.242.85.38]
  9  20 ms     30 ms    20 ms     fys-gw-xbb ldc.lu.se [130.235.8.111]
 10  20 ms     20 ms    30 ms     hermes.net.lu.se [130.235.128.100]

Trace complete.
D:\WINNT>
```

Nslookup/host

Med Nslookup, host, eller Dig kan man ställa frågor till DNS.



```
Command
D:\WINNT>
D:\WINNT>
D:\WINNT>
D:\WINNT>
D:\WINNT>
D:\WINNT>
D:\WINNT>nslookup ping.lu.se
Server:  barbafarfar.barbanet.local
Address:  10.0.12.1

Non-authoritative answer:
Name:     lundns.lu.se
Address:  130.235.128.100
Aliases:  ping.lu.se

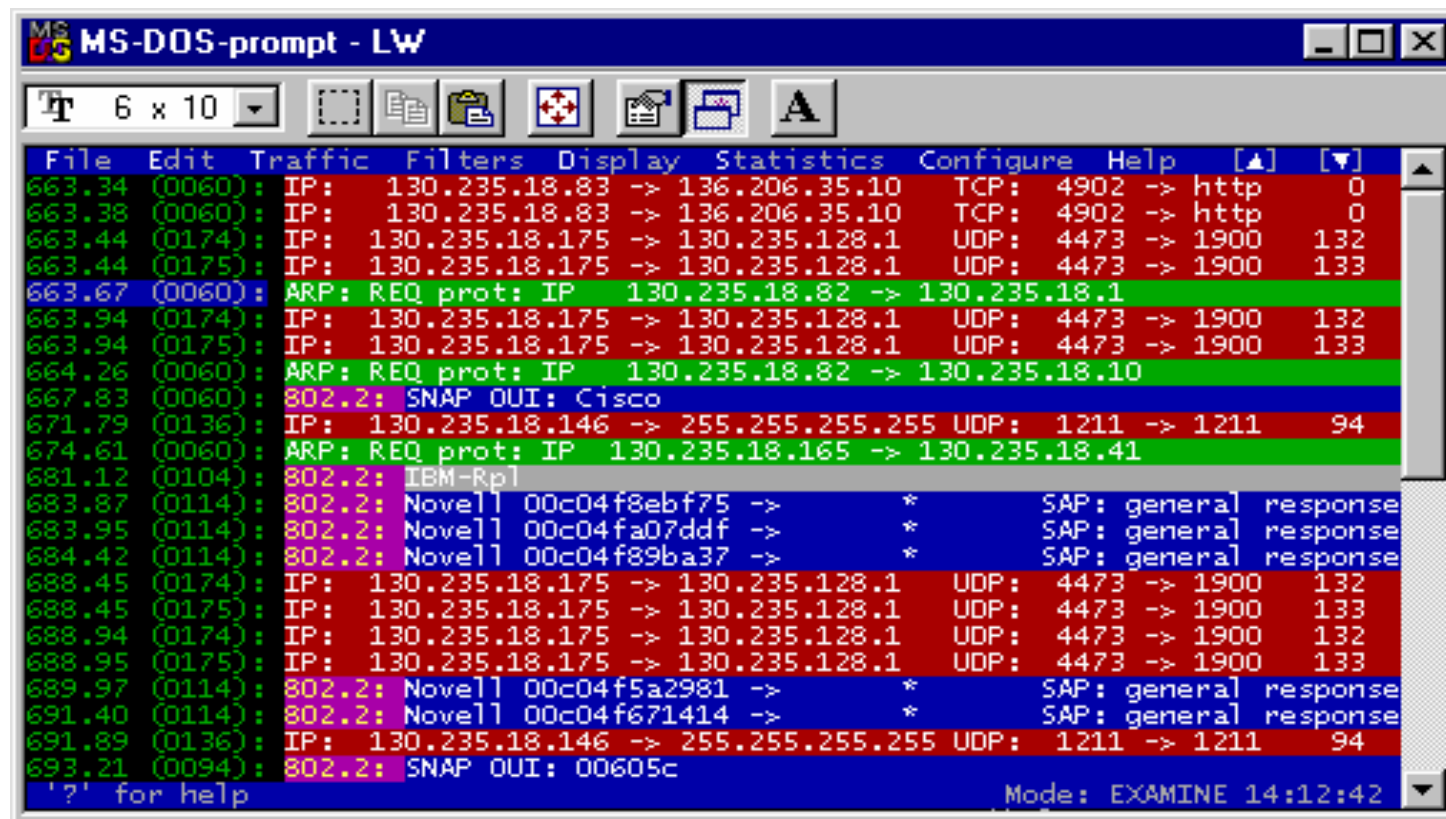
D:\WINNT>nslookup 130.235.128.100
Server:  barbafarfar.barbanet.local
Address:  10.0.12.1

Name:     hermes.net.lu.se
Address:  130.235.128.100

D:\WINNT>
```

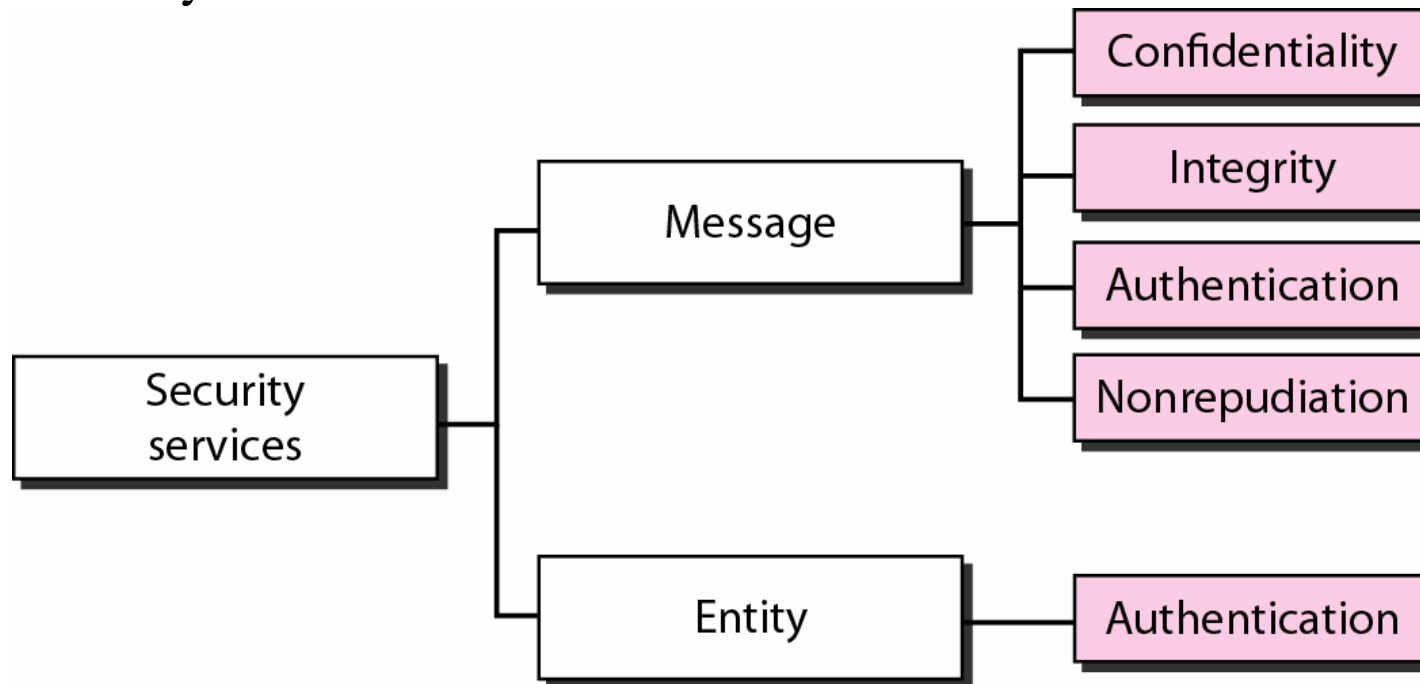
Sniffing

Some sniffer tools make it possible to “listen” to traffic on a link.



Network Security

There are five concepts that relates to network security:

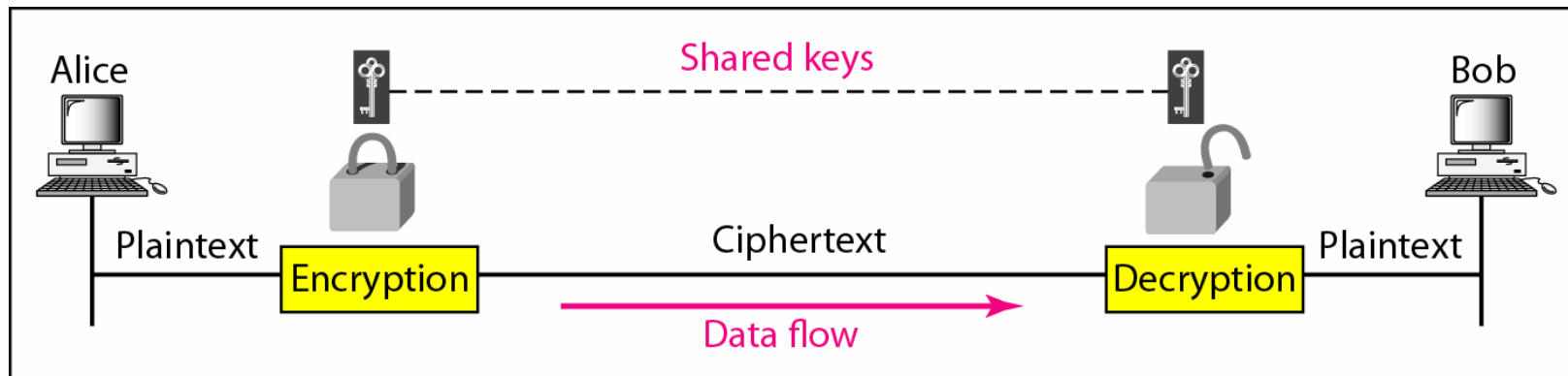


Message confidentiality (Privacy)

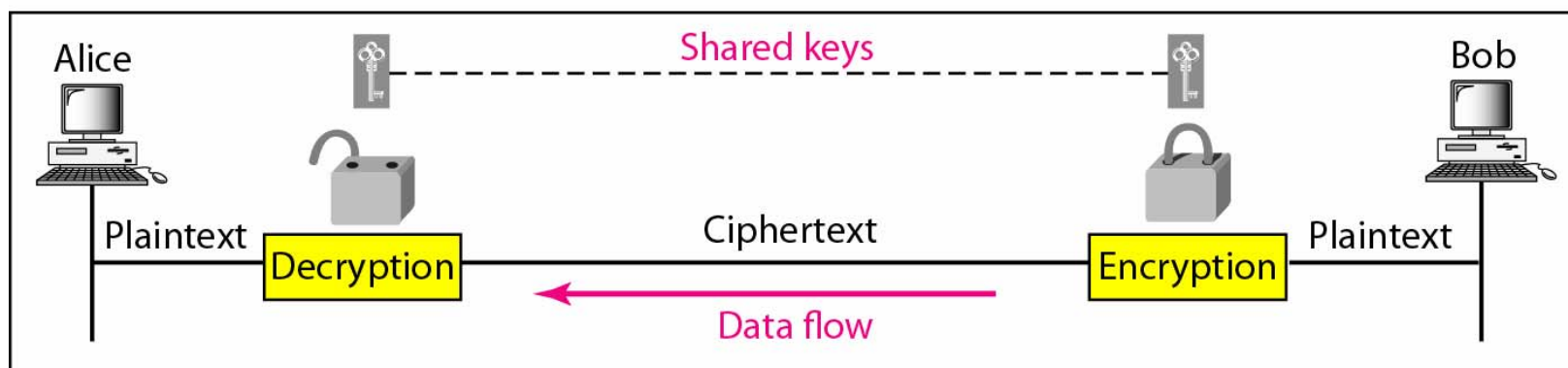
Message confidentiality (or privacy) means that the transmitted message must make sense to only the intended receiver. To all others, the message must be unreadable.

Privacy is usually achieved by encryption of the message.

Encryption, example



a. A shared secret key can be used in Alice-Bob communication



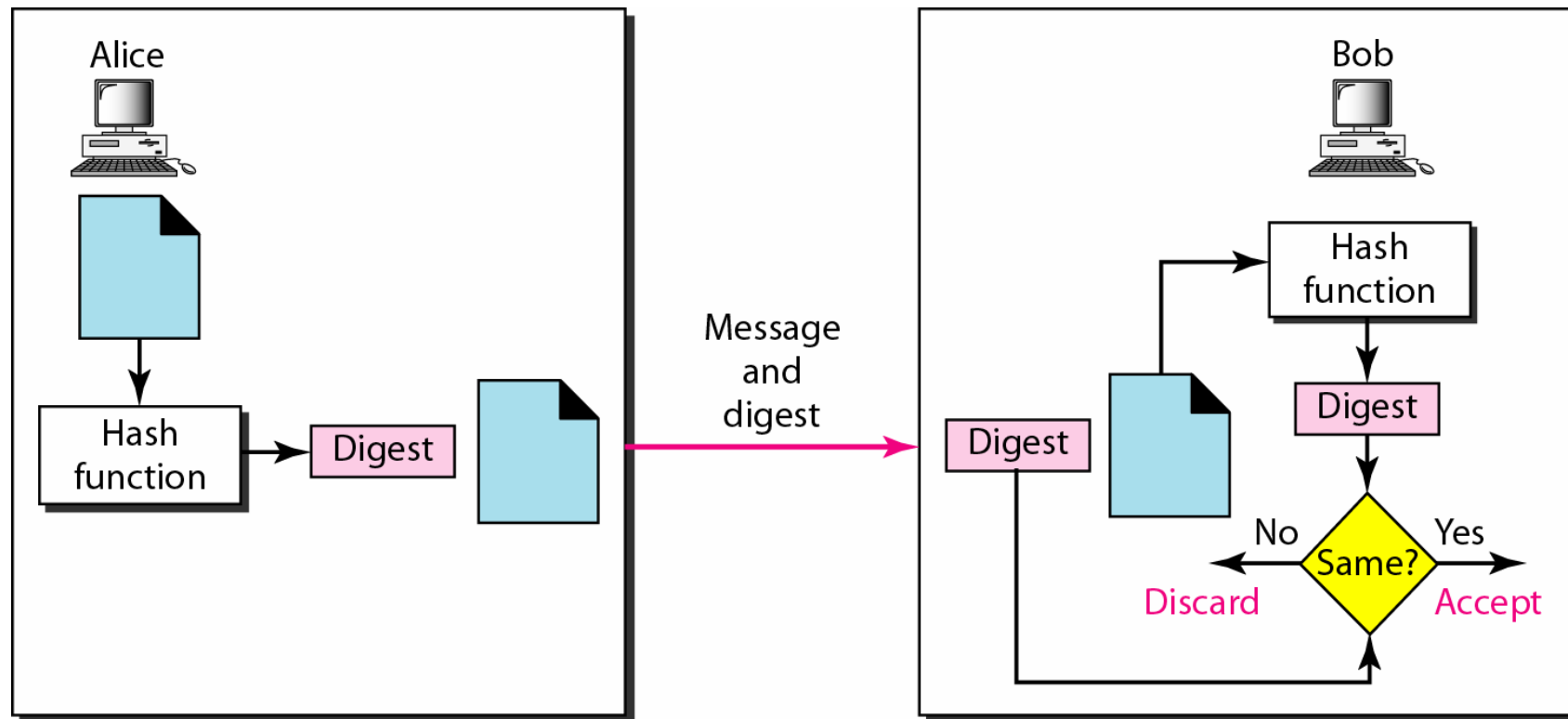
b. A different shared secret key is recommended in Bob-Alice communication

Message Integrity

Message Integrity means that the message must arrive at the receiver exactly as it was sent. There must be no changes during the transmission, neither accidentally nor maliciously.

Message Integrity can be achieved with [message digests](#).

Message digest

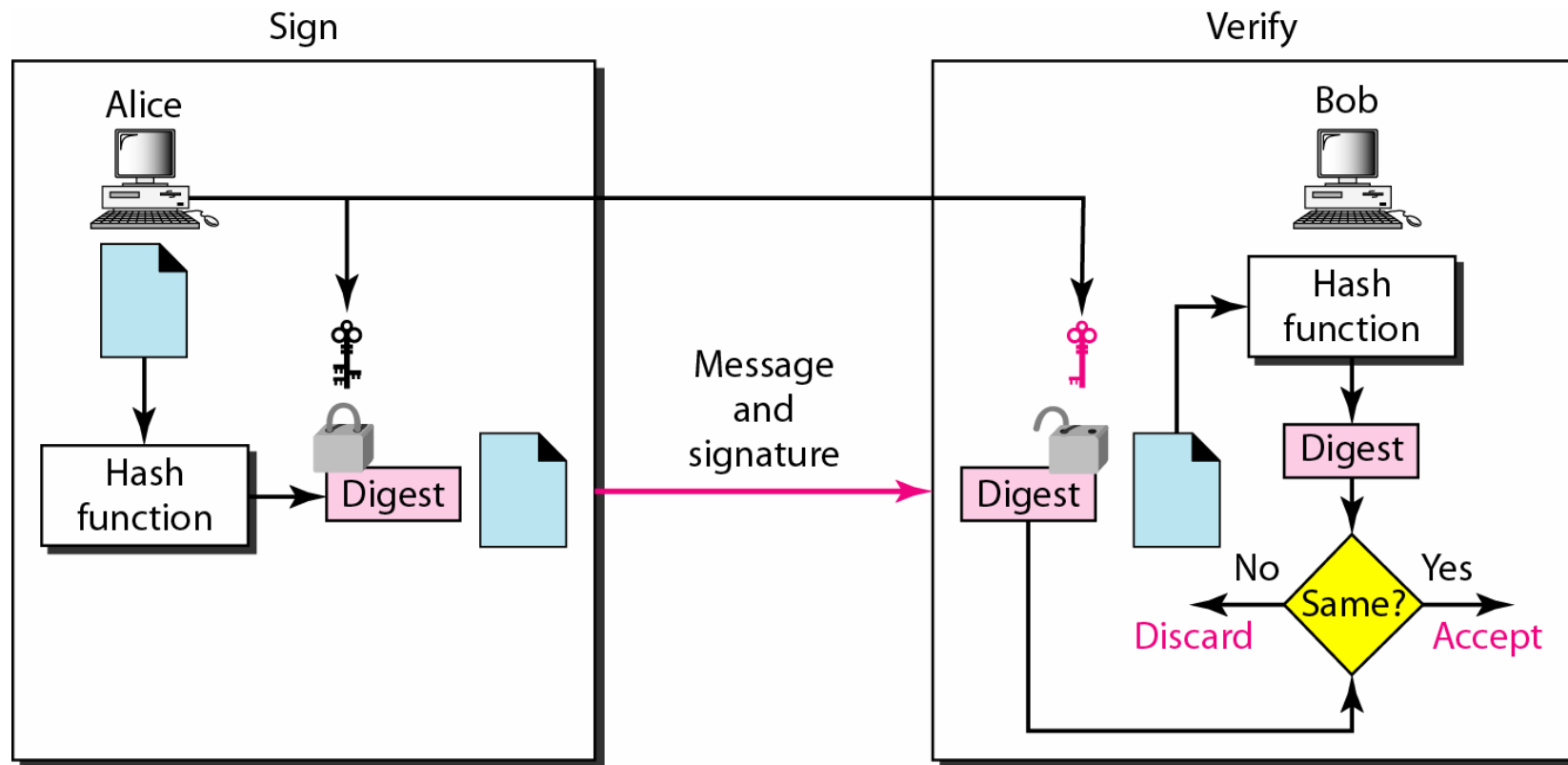


Message Authentication

In [message authentication](#) the receiver needs to be sure of the sender's identity and that an imposter has not sent the message.

Message authentication can be achieved with a keyed hash function that constructs an encrypted message digest, also called a [digital signature](#).

Digital signature

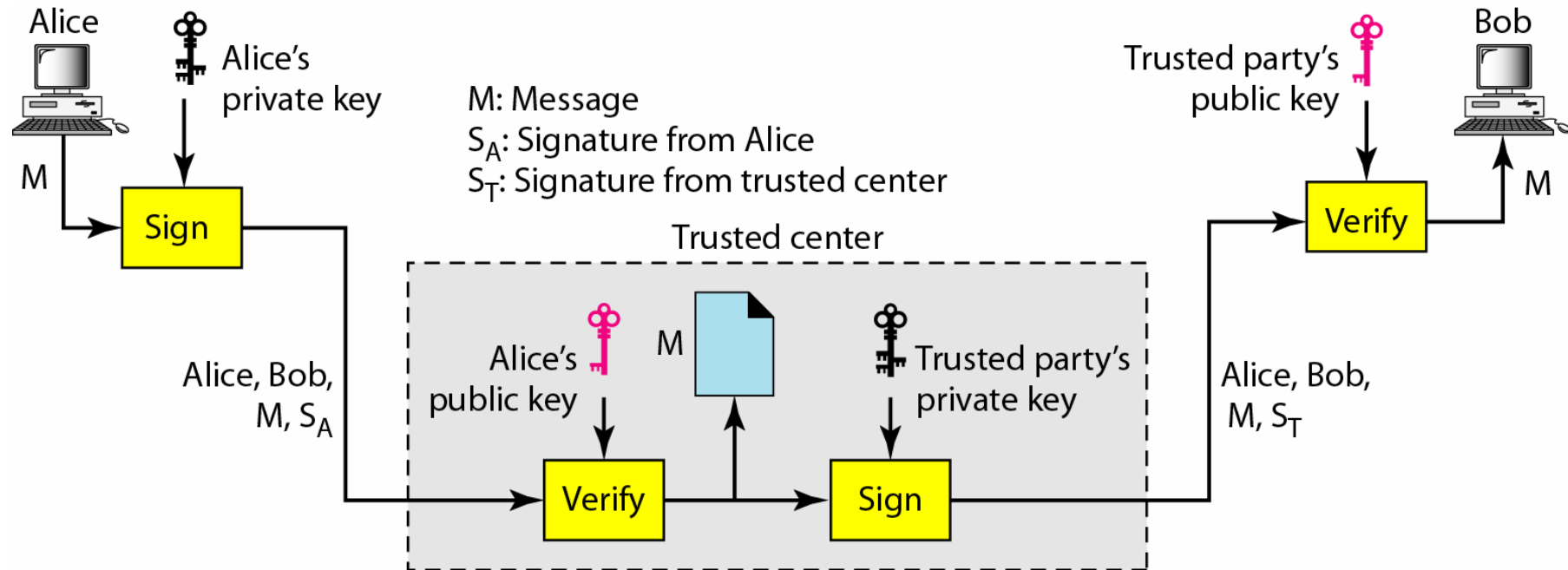


Message Nonrepudiation

Message nonrepudiation means that a sender must not be able to deny sending a message that he or she, in fact, did send.

Message nonrepudiation can be solved with a trusted third party and the use of digital signatures.

Trusted center for nonrepudiation

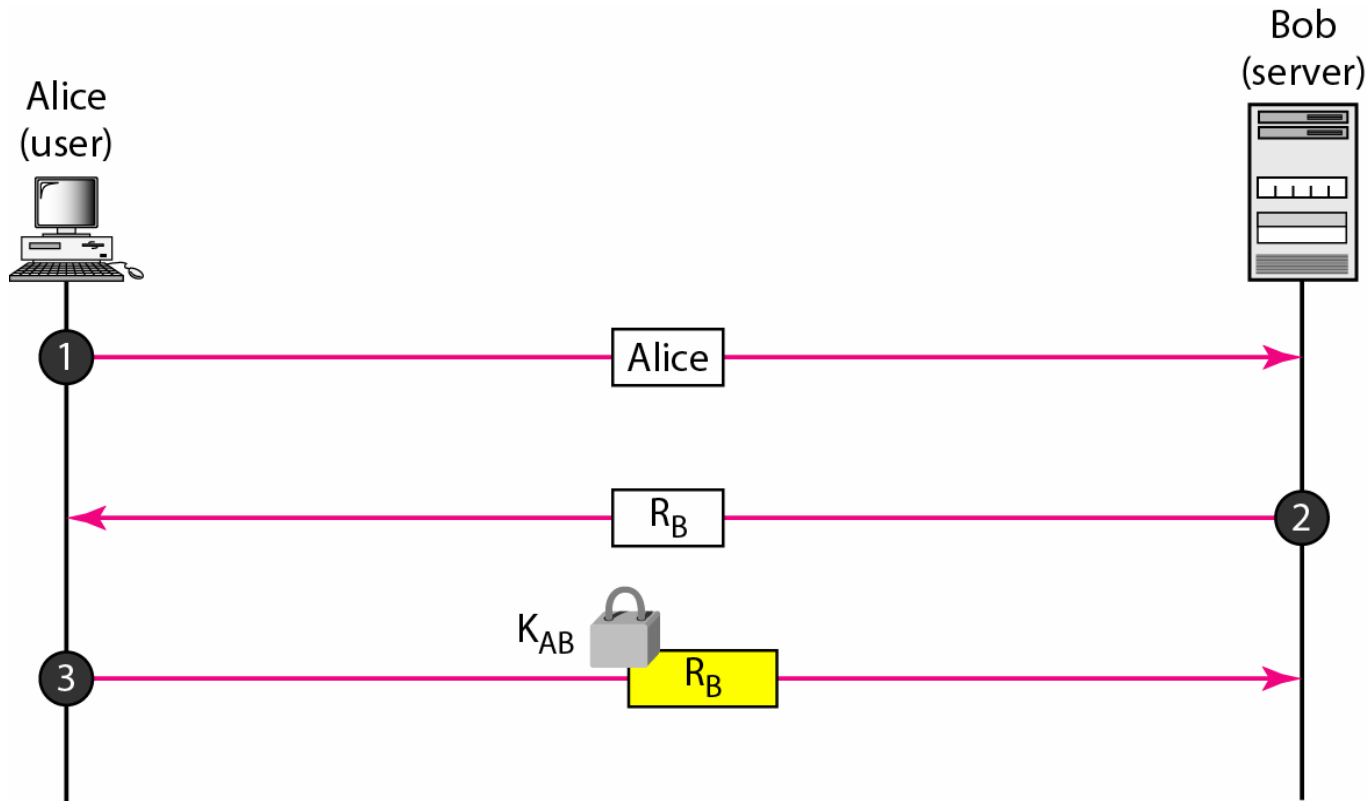


Entity Authentication

Entity authentication is a technique designed to let one party prove the identity of another party. An entity can be a person, a process, a client, or a server.

The simplest solution is to use passwords. Another solution is a so called challenge-response authentication.

Challenge-response authentication



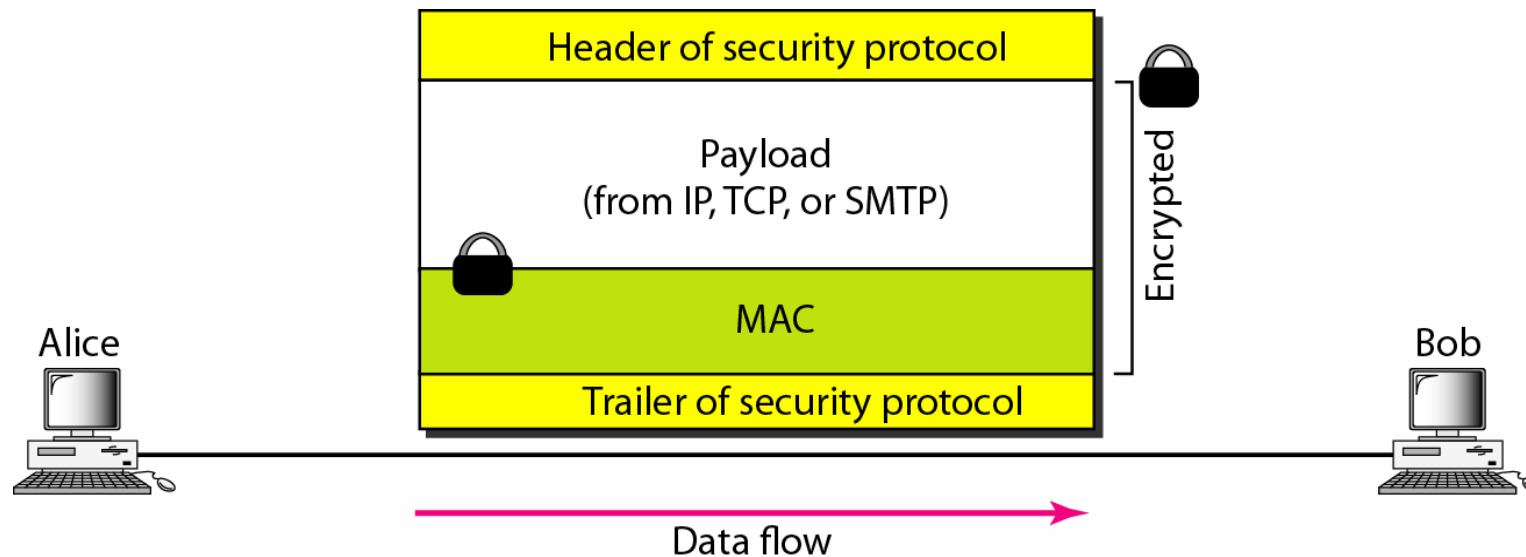
R_b = Random bit sequence (Nonce)

Internet security protocols

The following three security protocols ensure Authentication and Confidentiality on different layers:

- **IPSec**: Security for IP.
- **SSL/TLS**: Security for TCP.
- **PGP**: Security for Email (SMTP).

Common structure



MAC = Message Authentication Code (Digital Signature)

IPSecurity (IPSec)

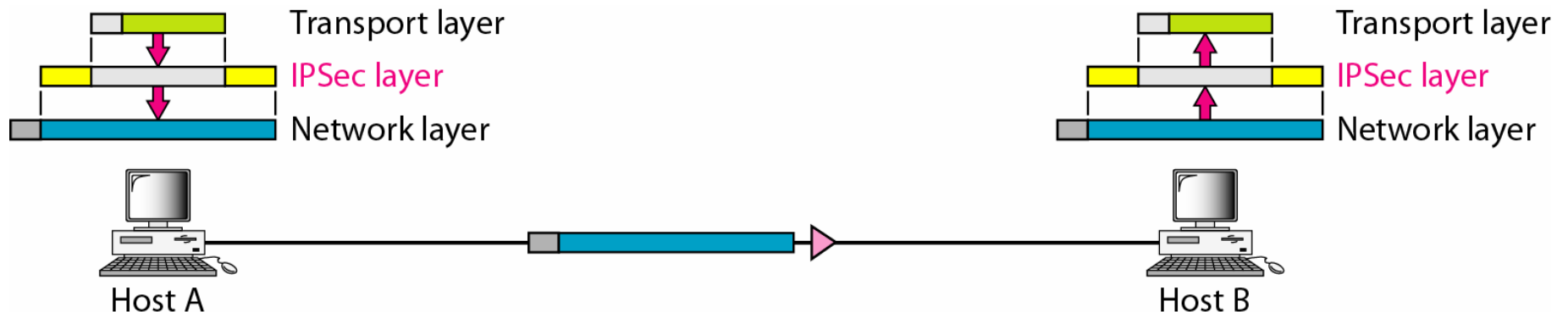
IPSecurity (IPSec) is a collection of protocols designed by the Internet Engineering Task Force (IETF) to provide security for a packet at the network level.

IPSec operates in one of two different modes:

- Transport mode
- Tunnel mode

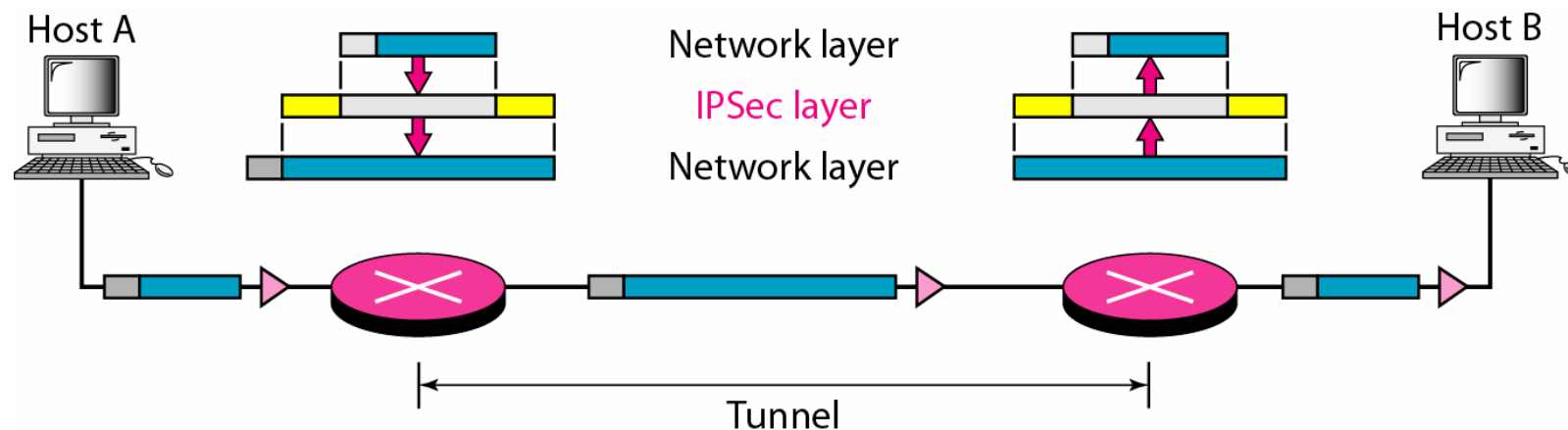
Transport mode

IPSec in the transport mode does not protect the IP header; it only protects the information coming from the transport layer.



Tunnel mode

IPSec in the Tunnel mode protects the entire IP packet. It takes an IP packet, including the header, applies IPSec security methods to the entire packet, and then adds a new IP header.

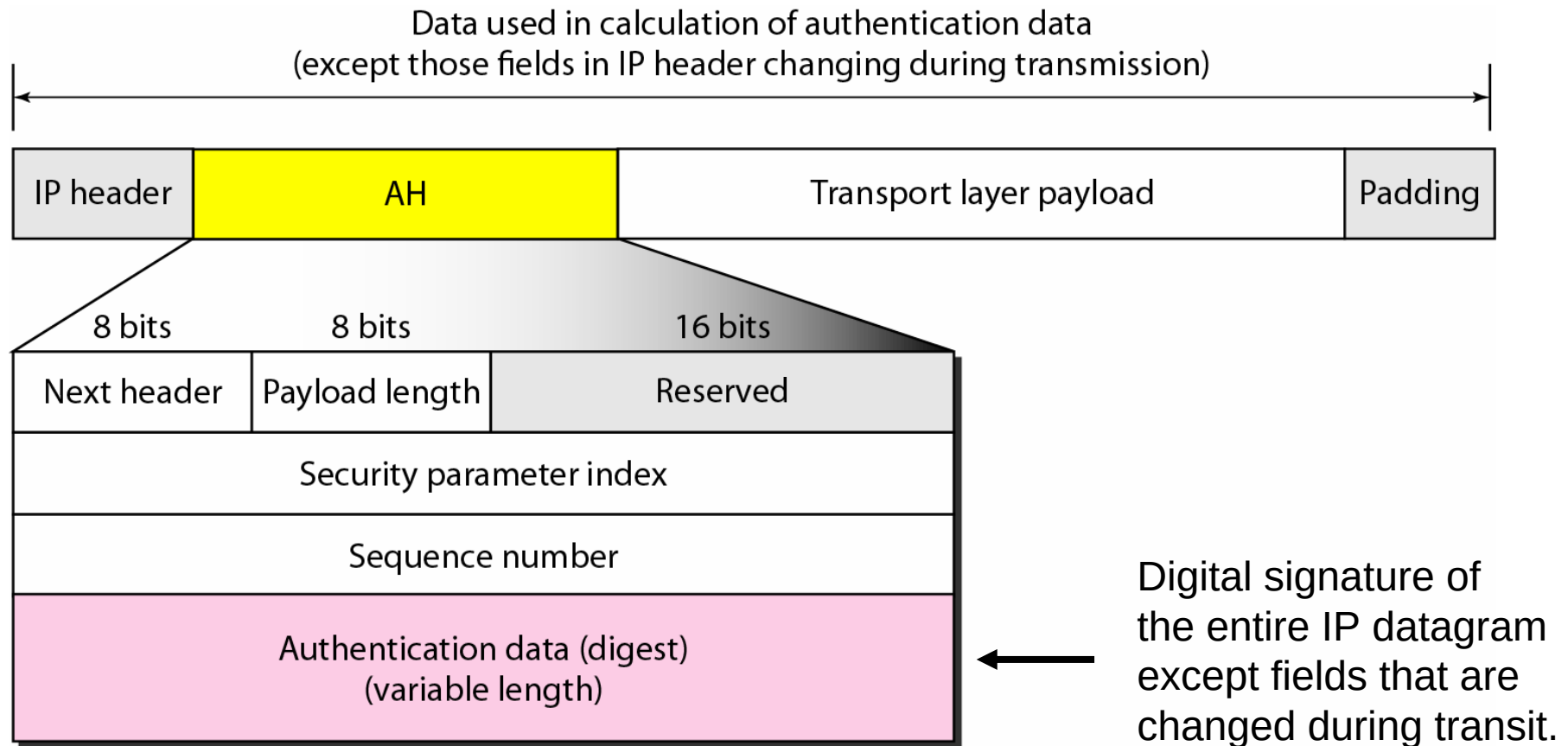


Two security protocols

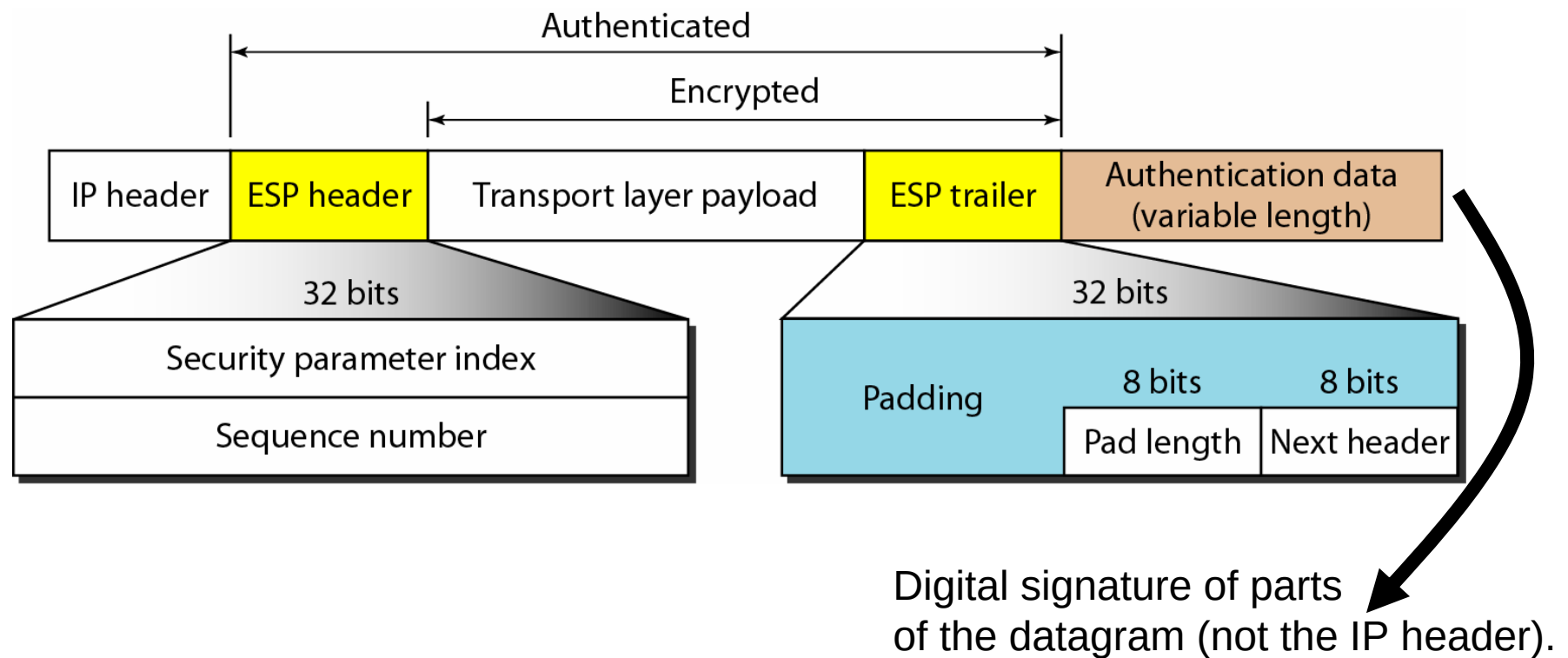
IPSec defines two protocols:

- Authentication Header (AH) Protocol
 - Provides source authentication and data integrity, but not privacy.
- Encapsulation Security Payload (ESP) Protocol
 - Provides source authentication, data integrity, and privacy.

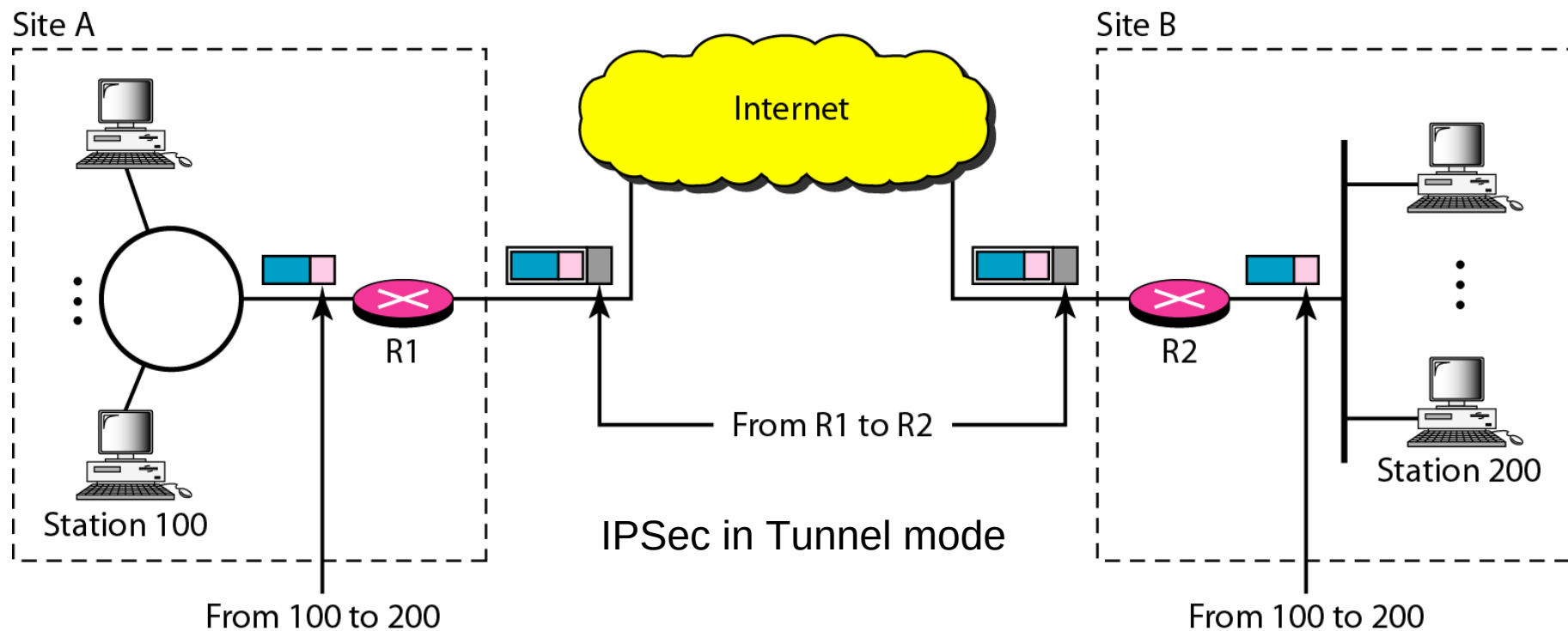
Authentication Header (in transport mode)



Encapsulating Security Payload (in transport mode)

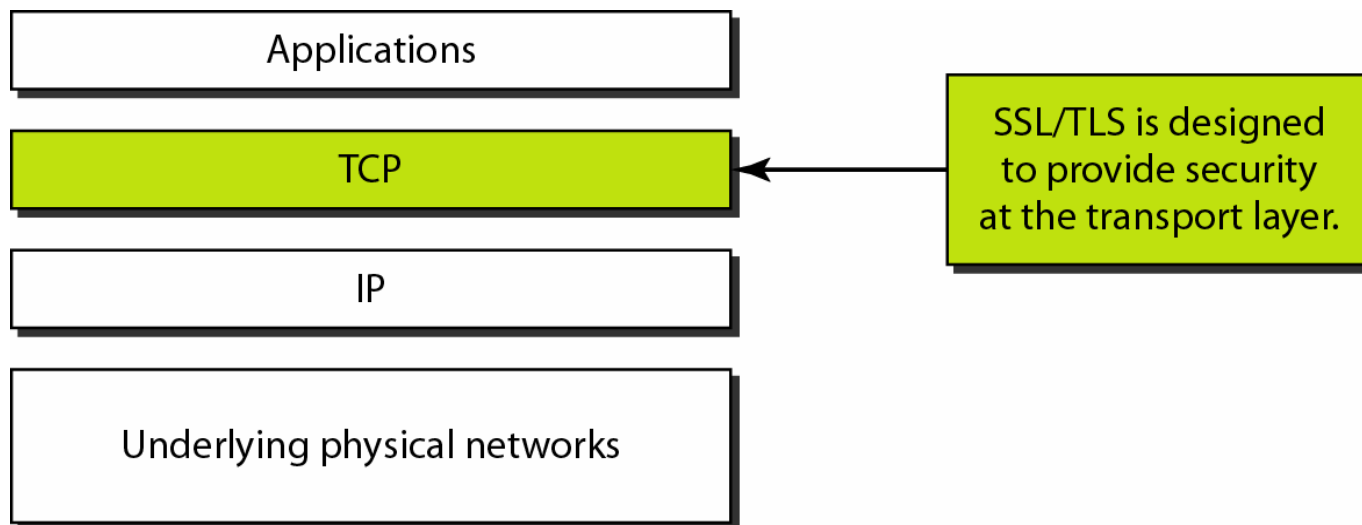


IPSec is used in Virtual Private Networks (VPNs)



SSL and TLS

Two protocols provide security at the transport layer: the [Secure Sockets Layer \(SSL\)](#) Protocol and the [Transport Layer Security \(TLS\)](#) Protocol. SSL was developed by Netscape, and TLS is an IETF version of SSL.

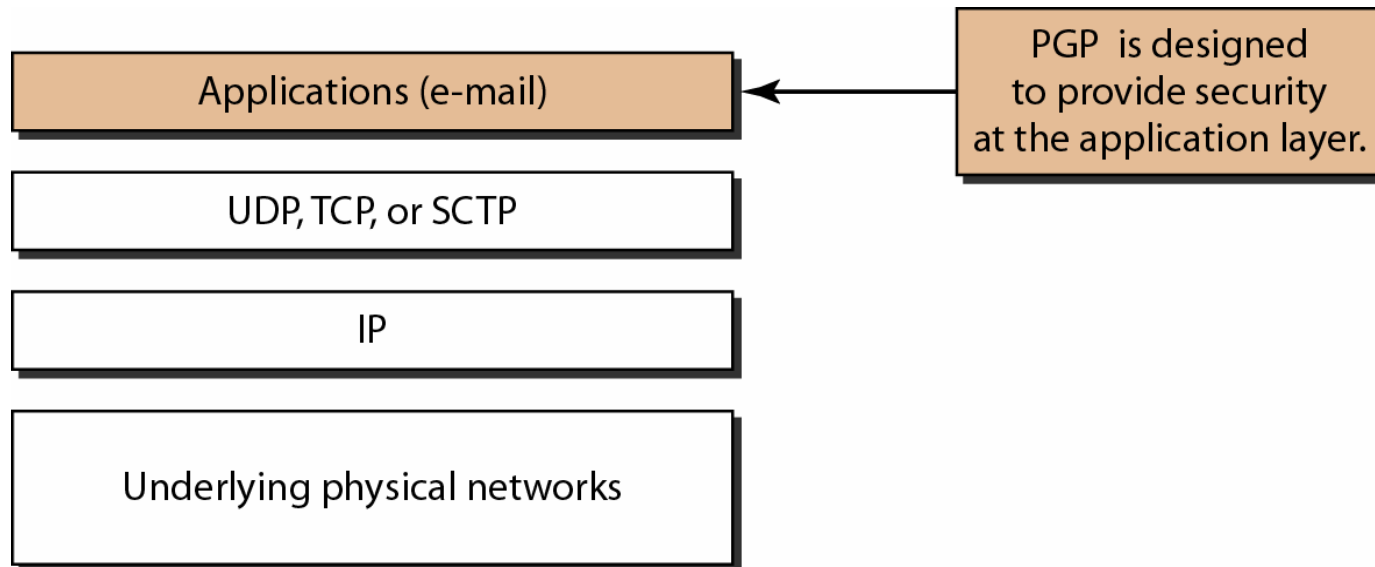


SSL/TLS services

- SSL/TLS provides **message integrity** and **confidentiality**.
- SSL/TLS can receive data from any application protocol, but usually the protocol is HTTP (https://x.y.z).
- The data is compressed, signed and encrypted.
- After this, the data is passed to a reliable transport layer protocol (TCP).

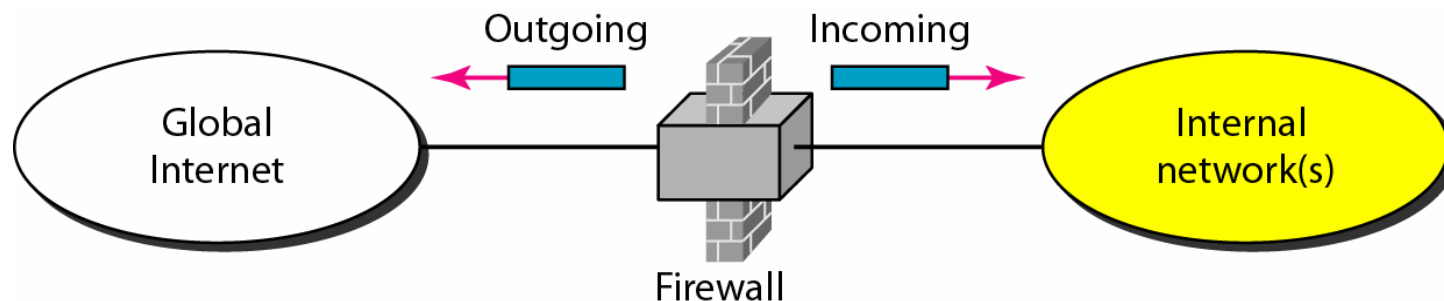
PGP (Pretty Good Privacy)

PGP was designed 1991 by Phil Zimmerman, and is today used to create authenticated and confidential emails.



Firewalls

A firewall is a device installed between the internal network of an organization and the rest of the Internet. It is designed to forward some packets and filter (not forward) others.



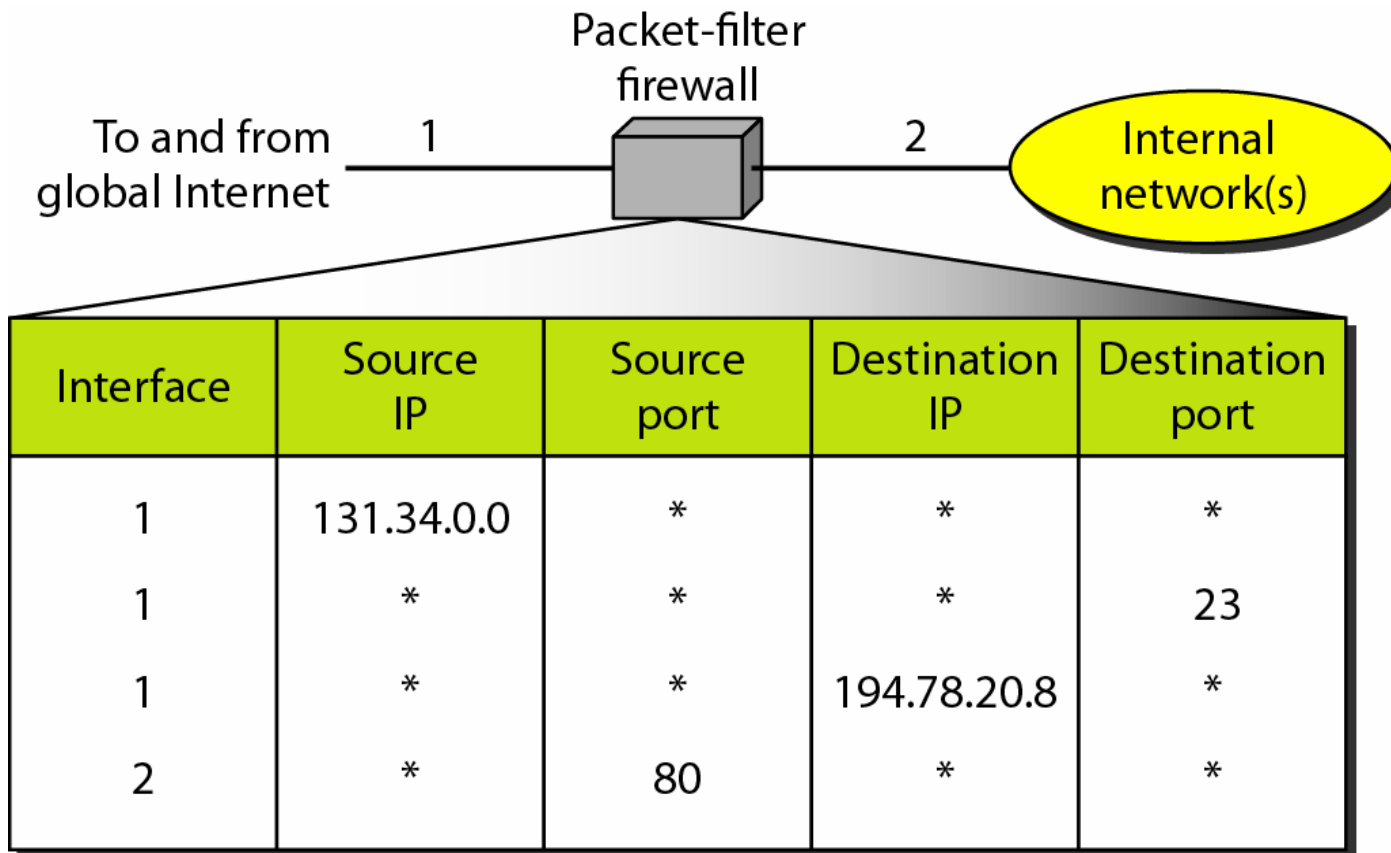
Packet-Filter Firewall

The **packet-filter firewall** works on the **network or transport layer**.

It can forward or block packets based on the information in the headers: source and destination addresses, port addresses, and type of protocol (TCP or UDP).

It is a router that uses a **filtering table** to decide which packets to forward or block.

A Packet-filter firewall



Proxy Firewall

A proxy firewall works on the application layer.

All messages go through a proxy HTTP server.

When an external client process sends a message, the proxy server must first accept it. If accepted, the proxy server then acts as a client process and sends the message to the real internal web server.

A proxy firewall

